

SPRAY DYNAMICS

SCARF FEEDER®

VIBRATORY CONTROLLER

This is a reference for setting up a new controller on a Spray Dynamics Scarf Feeder or Low Level Feeder. Please refer to the REO manual for information on how to scroll thru control parameters and sub menus.

Controller Setup Parameters (for a unit with an Accelerometer)

NOTE



**On all new equipment parameters will be preset
from Spray Dynamics - no changes are necessary**

Review and change the following parameters before tuning the vibratory feeder. Any other sub menus not noted do not need to be changed.

C003 -	E.S.P	0 - for consolidated control panel 1 - for standalone control panel
	4.20	0 - (not visible when E.S.P set to zero) 1 - for standalone control panel with pot
	Pot	0 - (not visible when E.S.P set to zero) 1 - for standalone control panel
	S.P.2.	0
	-En.	0
C008	A	100.00
	P	50.0 - varies to achieve max amplitude of tray
	F	46.8 - dependant on feeder natural frequency
	Acc.	1
	P.A.	40
	I.A.	100
	A.F.C.	1
A.F.S. DO NOT CHANGE OR SCROLL UP OR DOWN		
C020, 096	┐	1.0
	└	1.0

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Controller Setup Parameters (continued)

Change the Frequency Limit:

1. Go to control parameter C127, sub menu EnS – change EnS to 1
2. Go to C040, sub menu F.L. and change to 20
3. Go to C040, sub menu F.H. and change to 60
4. Go to C040, sub menu I: 33%

When all parameters have been adjusted go to C143, sub menu US.1 0 and push Δ until controller says safe. This will save all changes.

Autotune Controller

- Autotune the MFS268 controller by running the procedure detailed below (also reference the REOVIB MFS 268 manual)
 - Ensure that the tray is empty before starting the procedure and that nothing touches the tray while running the process.

*****NOTE*****



There are multiple sub menus that will be scrolled thru – only change noted parameters.

Go to Parameter C008 and adjust the following sub menus in order:

1. A - 000
2. P - 050
3. Enable Controller (Turn On)
4. F. – No Change
5. Acc. – No Change
6. P.A. – No Change
7. I.A. – No Change
8. A.F.C. – No Change
9. AFS – Push Δ - will find natural frequency in 1-3 minutes
10. P - 020
11. A - 100
12. P – Slowly increase until maximum amplitude ≤ 0.1 ” (read on sticker on the side of the vibratory tray) is achieved.

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Vibratory Feeder is now ready to be placed into service.

Reset Accelerometer Error

1. Make sure all items are plugged in and power is on.
2. Press the (F) key until the display reads (C000)
3. Press the (Up) Arrow key until display reads (C009)
4. Press the (P) or (F) key and the display should read (CL.err)
5. Press the (DOWN) Arrow key and the display should read (OFF)

The drive should read OFF. If the drive reads STOP, press the Green I key once and OFF should appear on the display. When turned on the feeder should read 100% on the display.